BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022 • Paper Code: PR X (01) 22

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** PR X (01) 22

Note: Four circles are given against each question. Fill only the circle of the correct answer. Use a blue or black marker/pen. Filling more than one circle will be treated as a wrong answer.

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. Sugars are polyhydroxy derivative of:

(A) Alkyl halides
(B) Aldehyde
(C) Ketone
(D) Aldehyde and Ketone

Q2. The coldest region in the atmosphere is:

(A) Troposphere
(B) Stratosphere
(C) Mesosphere
(D) Thermosphere

Q3. The density of water is maximum at:

(A) −40°C
(B) 40°C
(C) −4°C
(D) 4°C

Q4. All steps are used for the extraction of metals except:

(A) Roasting
(B) Smelting
(C) Evaporation
(D) Bessmiraization

Q5. Chlorine acts as bleaching agent in the presence of:

(A) HCl
(B) H_2O
(C) NaOH
(D) HNO_3

Q6. Percentage of nitrogen in urea is:

(A) 14 (B) 28

(C) 60 (D) 46

Q7. Salts formed when OH^- ions of a base are partially neutralized by an acid are called:

(A) Acidic salt (B) Neutral salt

(C) Basic salt (D) All these

Q8. Chemical reaction in which products can react again to form the reactants is called:

(A) Irreversible reaction (B) Reversible reaction

(C) Addition reaction (D) Substitution reaction

Q9. Which of the given is a Lewis base?

(A) HCl (B) $AlCl_3$

(C) BF_3 (D) F^-

Q10. Which of the given is an alcohol?

(A) $CH_3–O–CH_3$ (B) $CH_3–CH_2–OH$

(C) CH_3COOH (D) $CH_3–COH$

Q11. Dehydration of ethyl alcohol with conc. H_2SO_4 results in the formation of:

(A) Ethane (B) Methane

(C) Ethyne (D) Ethene

Q12. Carbon is bonded on both sides to oxygen atom in:

(A) Alcohol (B) Ketone

(C) Aldehyde (D) Ether

Section-B

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 32

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 45 minutes.

QII. Answer any EIGHT parts. Each part carries FOUR marks.

1. Describe Bronsted-Lowery concept of acids and bases.
2. Define the law of mass action.
3. Calculate the pH and pOH of 0.5 M solution of HCl. (pH = 0.301, pOH = 13.69)
4. How alkyl radicals are formed? Discuss with examples.
5. Which one is more reactive between alkane and alkene? Explain.

6. Differentiate between fats and oils.

7. What is acid rain? Explain.

8. Write any five physical properties of water.

9. Differentiate between mineral and ore.

10. Vitamins are vital for us, why?

11. Write any five different uses of urea.

Section-C

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 21

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 45 minutes.

Q. Attempt any THREE questions. All questions carry equal marks.

QIII.

(a) [4] How are proteins important to living organisms?

(b) [3] Distinguish between primary pollutants and secondary pollutants.

QIV.

(a) [5] What is salt? Write the different types of salts with examples.

(b) [2] State the conditions necessary for chemical equilibrium.

QV.

(a) [4] What are the different uses of organic compounds?

(b) [3] How alkenes are prepared by dehydration of alcohol?

QVI.

(a) [4] How blister copper is purified?

(b) [3] What are the causes of hardness of water?